

ExpenseEase Software Requirements Specification

Purpose and scope

ExpenseEase records shared group expenses, calculates balances, and proposes simplified settlement transfers. It supports local, single-currency INR use. Payment processing, OCR, mobile apps, notifications, recurring expenses, social login, and multi-currency support are out of scope.

Functional requirements

ID	Requirement	Priority
FR-01	Users register and authenticate securely.	Must
FR-02	Members create and manage groups.	Must
FR-03	Members add equal, exact, or percentage expenses.	Must
FR-04	The system computes balances from immutable records.	Must
FR-05	The system suggests deterministic simplified debts.	Must
FR-06	Authorized users record settlements.	Should

Non-functional requirements

Passwords use bcrypt with cost 10 or greater. Money is stored as integer paise and balance sums must equal zero. Inputs are validated before persistence. PostgreSQL runs in Docker for reproducible development and deployment.

Traceability

FR	User story	Test
FR-03	As a member, I want to split an expense fairly.	services.test.js
FR-04	As a member, I want accurate balances.	services.test.js
FR-05	As a member, I want fewer transfers.	services.test.js